Soumya Khaskel

Security Engineer | Product Security | Vulnerability Management | AWS & AI security
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WORK EXPERIENCE

INNER EYE CONSULTANCY SERVICES LIMITED **JUNE 2026 - July 2026**
DevSecOps And Network Security Intern (LPU MCA Mandatory Training) ~ Kolkata, INDIA
~ Architected and build a **5 stage CI/CD pipeline** via **Github Actions** with a 5 branch promotion model enforcing manual **approval gates** after **security** checks and at each **environment**, integrated **OWASP Dependency scanner** with **NVD API** and then to **Trivy** to cut **30+ minutes testing time to 30 seconds** via **DB caching**, with **zero loss** of coverage. Integrated **3 layer automated** security scanning(**Gitleaks** for **secrets**, **Trivy** for Dependency and **Semgrep** for **SAST**) while building a **python aggregation script** to compute a weighted **0-100 security score** from **scan results**.
~ Discovered and **remediated** real disclosed **CVE's** (**Spring Boot Actuator** auth bypass, **Tomcat/Jackson-databind vulnerabilities**) surfaced by the pipeline itself, improving the security score to **100 after remediation** also **Provisioned** and **hardened** a **production Linux Server (AWS EC2)** with **systemd** managed services, **NGINX reverse proxy** and **least privilege SSH** sudo access. **Automating zero touch deployment** via SSH with retry based **health checks** and **symlink** based **rollback** capability.
~ Architected an end to end **threat modeling framework** for our organization **identifying 39** actionable **threat** scenarios covering **Web applications**, Zoho **SAAS**, network infrastructure, physical security and third party risk register with **mitigation** tracker to **7 day containment** plan for **high risk** area and executive summary tailored for **SME** environments considering **ISO/IEC 27001 Security Controls**.Making quick **report documentation** with GENAI tools.

TATA CONSULTANCY SERVICES **MAY 2025 - MARCH 2026**
Assistant System Engineer ~ Chennai & Kolkata, INDIA
~ Delivered L1 **security operations** support handling **phishing**, **identity/access** and **endpoint threat** tickets (20-30/day) using security tools like **Microsoft Entra ID**, **Intune Portal**, **CrowdStrike Falcon** and **DLP** tooling. Escalated a critical **RDP vulnerability** that **triggered** a project wide **security patch upgrade**. Typically closing **97%** within SLA.
~ Identified a gap in **trainee assessment** created by widespread **GENAI** tool use. Led a team to **build** a **VS Code extension** that tracked coding **activity signals** in real time to feed a **JavaScript** based **dashboard** for the **SMEs** used to evaluate **GenAI assistance** levels across a **40 person training cohort**. Recognized with **TCS Ignite Values Champion** award for training **performance** in Chennai.

PROJECTS - [Github_Link]

LLM Prompt Injection Detection Tool - Python, AWS Bedrock, boto3, YAML
~ Identified the lack of structured, repeatable tooling for testing LLM safety boundaries as a gap in practical AI security workflows, and built a local first security testing engine that executes 53 adversarial prompt injection attacks across 10 categories including jailbreaks, delimiter injection, authority impersonation, and encoded payload smuggling against foundation models via Amazon Bedrock. Evaluated **Meta Llama 3.3 70B** revealing a 64% failure rate with critical findings including full Developer Mode activation and emergency override acceptance, generating timestamped JSON security assessment reports per scan. **[View]**

AI CyberSecurity_News_Agent - Python, FastAPI, SQLite, Groq API(Llama 3.1), APScheduler, Telegram BotAPI
~ Engineered and deployed an autonomous AI agent that aggregates security news from sources like (CISA, TheHackerNews, Krebs, BleepingComputer)every 30 minutes, using LLM's like Groq's Llama 3.1 8B (gemini in backup)to auto summarize and severity tagging each article with SHA-256 based deduplication. Delivering real time mobile alerts via Telegram so critical threats surface before opening a laptop. [**View**]

Security Audit Dashboard - Python, YAML, Bash/Shell, Spring Boot, React, PostgreSQL
~ A live security observability tool that ingests results from an automated CI/CD security pipeline and displays a real time, weighted security score, findings breakdown, and deployment history for every code change replacing manual security reporting with a single, always current dashboard deployed in a AWS EC2 instance for 24/7 checks. [**View**]

Distributed Infrastructure Monitoring Lab - Zabbix, Docker & Cross-ISP VPN Mesh, Bash/PowerShell
~ Built a self hosted Zabbix Monitoring stack with Docker compose, MySQL, frontend on macOS to monitor 4+ endpoints across a Windows PC and geographically distributed endpoints on independent ISPs solving cross NAT connectivity by evaluating and rejecting multiple VPNs before implementing TailScale mesh VPN for stable, routable connections across independent ISP's. **[View]**

Network Performance Optimization - QoS, DNS & Protocol Level tuning for competitive gaming
~ Conducted a systematic NOC style audit of a home network (ISP - router) to eliminate latency spikes impacting competitive FPS gameplay, diagnosing QoS misconfiguration as root cause through packet level analysis. Implemented DSCP 46 traffic prioritization via Windows Registry, migrated DNS resolution to Cloudflare, and tuned TCP/IP stack and MTU settings reducing average spikes 192ms. Documented the full methodology, CLI commands, and before/after packet capture evidence in a published technical report, sharing findings with the gaming and networking community via GitHub, Dev.to, Medium and LinkedIn. **[View]**

SKILLS

Security Engineering & DevSecOps: SAST/SCA Security Scanning, Semgrep, Trivy, Gitleaks, OWASP Dependency-Check, NVD API, CVE Remediation, CWE & CVSS Analysis, CI/CD Security Gates, Secrets Detection, Threat Modeling, Risk Register Development, ISO/IEC 27001, Vulnerability Management, Penetration Testing, Nmap, Nessus, Burp Suite, AI/LLM Security, Metasploit, OpenVAS, Prompt Injection & Jailbreaking Defense, Exploitation. GitHub Actions, Branch Promotion Workflows, Approval Gate Design, Pipeline Automation, Health-Check Automation.

AI & LLM Security: Agentic AI security, LLM security, Prompt Engineering Based Automation, Agentic AI Architecture & Orchestration, LLM API Integration (Groq/Llama, Gemini,Claude & GPT), Context Management, AI Workflow Automation, Autonomous Agent Development, MCP Integration, AI Driven Severity Scoring, TryHackMe AI Security Certified Prompt Injection, Jailbreaking, OWASP LLM TOP 10 & Agentic Top 10, hands-on labs. AI-Assisted Analysis.

SOC & Security Incident Response: Alert Triage, Phishing Investigation, Identity & Access Management, Endpoint Threat Detection, Microsoft Entra ID,SIEM, Microsoft Intune, CrowdStrike Falcon, DLP, RDP Vulnerability Remediation.

Cloud & Infrastructure Automation: AWS, EC2, IAM, IAC, Terraform, Linux Server Hardening, IAM policies & least privilege, systemd, NGINX Reverse Proxy, SSH Least-Privilege Access, Zero-Touch Deployment, Automated Rollback, Infrastructure Monitoring, Zabbix, Docker, Docker Compose, VPC, Web Application Firewall,Cloud Architecture & System Design, MCP Integration.

Programming & Backend Basics: Python, FastAPI, Spring Boot,, PostgreSQL, SQLite, Rust, YAML, REST API Development, HTML, CSS, JavaScript, Bash/Shell Scripting, SSH, PowerShell

Tools & Platforms: Git & GitHub, Telegram Bot API, APScheduler, Systemd, Linux, Windows Registry Configuration, macOS/Windows Cross-Platform Administration.

CERTIFICATIONS and ACHIEVEMENTS - [Github_Link]

Certified Ethical Hacker (CEH) — EC-Council (via Simplilearn)
Microsoft Certified: Azure Security Engineer Associate (AZ-500) — Microsoft (via Coursera)
AWS Certified AI Practitioner — Amazon Web Services (In Progress)
Claude Certified Architect (CCAR-F Foundations) — In Progress
TryHackMe AI Security Certification (AI1) — TryHackMe, May 2026
TryHackMe Advent of Cyber 2024 — TryHackMe
Google Cybersecurity Professional Certificate — Google
Network Forensics & PCAP Analysis — Security Blue Team
OSINT for Defenders — Security Blue Team
Dark Web Operations & Threat Intelligence Collection — Security Blue Team
Data Structures & Algorithms (Java) + MERN Stack Development — Apna College
AWS Skill Builder — Cloud Foundations & Digital Badges.
TryHackMe Top 4% & TCS IGNITE VALUES CHAMPION

EDUCATION

Maulana Abul Kalam Azad University of Technology **2021 – 2024**
Bachelor of Computer Application (BCA - 8.5/10) ~ Kolkata, West Bengal, INDIA

Lovely Professional University **2024 - 2026**
Masters of Computer Application in Cyber Security (Completed results awaited) ~ Phagwara, Punjab, INDIA